

Combining the Fine-Tuned LLM with the Audit Agent

Two systems were built to do the same job — flag non-inclusive language in academic text. They sit at **opposite ends of the precision / recall tradeoff**. This is the integration plan that turns that contrast into a single cascade.

Models · Qwen2.5-3B QLoRA (r8_d0.2_achva) + Gemma-BGE-M3-Qdrant agent **Date** · 2026-06-21

The headline: the **fine-tune catches more but over-flags**; the **agent is near-perfect on precision but misses half**. Cascade them — fine-tune as the recall generator, agent as the precision gate — and each covers the other's largest gap. Two exceptions are called out honestly: Hebrew (one-directional fill) and Outdated (a shared blind spot).

1 The two profiles

agent: Achva-EN 50 · fine-tune: expert-78 / gold-65

Audit Agent **PRECISION SPECIALIST**

LangGraph ReAct + Reflection + Agentic-RAG · Gemma + BGE-M3 + Qdrant

Precision (Achva-EN)	1.00
Recall	0.43
F1	0.60
False positives	0 / 10

- + Cited RAG grounding + ask / retract control paths
- + Never flags clean text — safe to surface to users
- Misses >half of real violations (recall 0.43)
- Hebrew not supported (regex sentence splitter)

Fine-Tuned LLM **RECALL SPECIALIST**

Qwen2.5-3B-Instruct · QLoRA r8, dropout 0.2 · single forward pass

Expert strict — macro-F1	0.706
Expert — use-mention FP	0.118
Gold (real docs) — FP	0.607
Hebrew F1 (expert / gold)	0.559 / 0.338

- + Strong class coverage: Pot-Offensive F1 0.84, Biased 0.69
- + Handles Hebrew · cheap single-pass inference
- Over-flags ~60% of clean spans on real prose
- No grounding / citation for its calls

Why they compose — opposite ends of one axis

HIGH RECALL · low precision

AG

Agent

HIGH PRECISION · low recall

FT

Fine-tune

A high-recall generator feeding a high-precision verifier is the textbook cascade: the fine-tune surfaces candidates (lifts recall), the agent's `classify_span` + RAG + retract path rejects the bad ones (lifts precision). Neither is shippable alone — together they plausibly beat both.

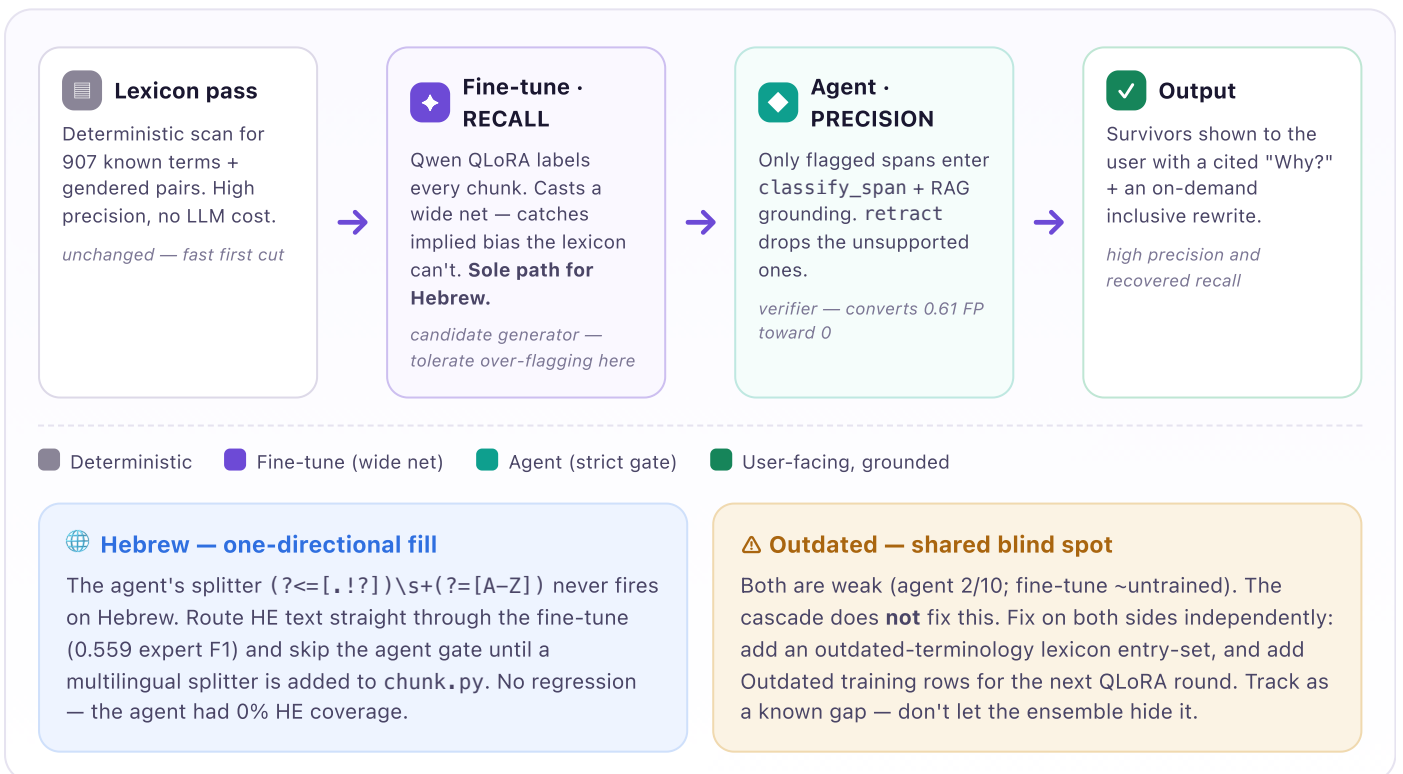
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✦ winner per dimension

Dimension	Audit Agent	Fine-Tuned LLM	Who covers the gap
Precision on real docs	1.00 — 0 FP	0.607 FP (gold)	AGENT verifies the fine-tune's flags → kills its false positives
Recall / catch rate	0.43 — misses half	Pot-Off 0.84 · Biased 0.69 F1	FINE-TUNE generates more candidates → lifts the agent's recall
Hebrew	unsupported (splitter)	0.559 / 0.338 F1	FINE-TUNE is the sole HE path until a multilingual splitter lands
Grounding / "Why?"	cited ERIC RAG	none	AGENT supplies the citation + explanation layer
Cost / latency	multi-hop + vector DB	one forward pass	FINE-TUNE runs on everything; agent runs only on flags
Outdated category	2 / 10 caught	~untrained (0 strict)	NEITHER — shared blind spot, fix both sides separately

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maps onto the documented lexicon → LLM → RAG pipeline



4 Wiring it to the agent repo

inclusify-audit-agent · offline-first contract

- 1 **Serve the fine-tune** behind the agent's existing OpenAICompatLLM interface (vLLM on the T4 VM). No new code path — it slots in where Gemma sits today via `.env`.
- 2 **Add a "candidate" mode**: the fine-tune runs first over all chunks and emits flagged spans. The agent's act node consumes them instead of starting from the lexicon miss.
- 3 **Gate, don't re-detect**: each fine-tune flag is escalated to `classify_span + retrieve_citation`. Keep only spans the agent confirms *and* grounds.
- 4 **Route by language**: HE chunks bypass the agent gate (fine-tune only) until the splitter is multilingual.
- 5 **Confidence wiring**: low-grounding flags hit the tuned retract path (high $\geq$ 0.5 / low $<$ 0.35). This is the FP killer on real prose.

```
# span contract: fine-tune → agent gate
{
  "text": "...he is the chairman...",
  "lang": "en", # he → bypass gate
  "span": [12, 20],
  "label": "Biased",
  "source": "finetune" # recall layer
}

# agent verifies → final verdict
if agent.classify_span(s).confirmed
  and citation.score >= 0.5:
  keep(s) # grounded → show user
else:
  retract(s) # kills the FP
```

Honors the repo's offline-first rule: the fine-tune sits behind a provider interface; MockLLM still works with no keys.

5 Prove it before trusting it

the one missing measurement

The complementarity is currently inferred, not measured. The agent was scored on Achva-EN (50), the fine-tune on overlapping-but-different expert (78) and gold (65) sets. Before relying on the cascade, run **both systems + the combined pipeline on one identical Achva gold set** (50 EN now; +50 HE once the splitter lands) and measure end-to-end precision / recall / F1.

Expected outcome (hypothesis)

- Recall > **0.43** (agent alone) — fine-tune adds candidates
- Precision » **0.39** (fine-tune alone on gold) — agent gate filters
- Hebrew coverage **0% → ~0.34–0.56 F1** via fine-tune path
- Outdated stays weak — confirms the documented gap

Action checklist

- Freeze one shared Achva gold set (EN + HE) ~½ day
- Serve fine-tune via OpenAICompatLLM ~½ day
- Add candidate-mode + lang routing to act ~1 day
- Multilingual splitter in `chunk.py` ~½ day
- Outdated lexicon set + training rows later round